DBMS Term Project – Student Reflection Form

Section 1: Student Information

Name: Wisit Suwannao

Student ID: 67070501042

Group No: 6

Group Name: มูลนิธิกันจอมพลังช่วยสู้

Project Topic: Modlearn Streaming Education Platforms

Section 2: Peer Recognition

🏆 Best Presentation (Other Teams)
Which team gave the most effective or impressive presentation?
Group No: 1
Reason:

This part of the project stands out because it's the only one that focuses on a music streaming platform.

👤 Best Project Member (Your Team)
Which teammate contributed the most to your project's success?
Name: Supawit Marayat (67070501045)
Reason:

He professionally handled the backend and database system design, delivering a highly efficient architecture. His contributions were instrumental in helping the whole team align and streamline our frontend and backend development plans.

Section3: Project Reflection

1. Project Summary

Briefly summarize your term project:

- What was the project about?

The project was about building ModLearn, a modular online learning platform for our CPE241 Database Systems final project, where we applied database design and full-stack development to create a system that supports user authentication, course and lesson management, category organization, personal libraries, watch progress tracking, reviews, recommendations, analytics, and commerce-related features in one integrated application.

- What were the main features or goals?

The main features and goals of the project were to create a complete online learning system that could manage courses and lessons, handle user accounts and authentication, organize content by categories, track each user's learning progress, support reviews and recommendations, manage files and media, provide analytics, and demonstrate a well structured database-backed application that solves real-world learning platform needs.

2. Technical Learning

What DBMS-related skills and knowledge did you develop?

(Check all that apply and explain.)

- ☑ ER Modeling
- ☑ Normalization
- ☑ SQL (DML/DDL)
- ☑ Transactions & Concurrency
- ☑ Indexing & Optimization
- ☐ NoSQL Concepts
- ☑ DBMS Software (e.g., MySQL, PostgreSQL)
- ☐ Others: __________

Explanation:

this project, I developed ER modeling skills by identifying key entities such as users, courses, lessons, categories, reviews, purchases, and watch progress, then defining the relationships between them clearly. I also applied normalization to structure the database properly, reduce redundancy, and maintain data consistency. My SQL skills improved through creating tables, defining constraints and relationships, and performing database operations for the system. I also gained understanding of transactions and concurrency by recognizing the need for reliable multi-step operations and consistent updates when handling user activity, purchases, and progress data. In addition, I developed knowledge of indexing and optimization by understanding how indexes help improve query performance for frequently accessed data such as courses, users, and progress records. The project also gave me practical experience using PostgreSQL as the DBMS and integrating the database with the backend in a real application.

3. Project Challenges and Solutions

Describe one major challenge your team faced and how you resolved it. One major challenge our team faced was designing a database structure that could handle many related features at once, such as courses, lessons, categories, reviews, purchases, and watch progress, without making the system messy or redundant. We resolved this by carefully planning the entity relationships, normalizing the tables, and refining the schema step by step so that each feature had a clear connection in the database. This made the system more organized, easier to maintain, and more reliable when integrating it with the backend and application logic.

4. Teamwork Reflection

How was the collaboration in your team?
- What worked well?
- What could have been improved?

The collaboration in our team worked well because we were able to divide tasks across different parts of the project, such as database design, backend development, and other system features, which made the workload more manageable. Using shared tools like GitHub also helped us coordinate changes and keep track of progress. What could have been improved was communication and early planning, especially in making sure everyone had the same understanding of the database structure and feature connections before implementation, which would have reduced confusion and rework later on.

5. Tool and Technology Use

Which tools or platforms did your team use, and how did they support the project?
(e.g., GitHub, DBMS software, project management tools)

Our team used GitHub for version control and collaboration, which helped us manage code changes and work on the project together more efficiently. We used PostgreSQL as the DBMS to store and manage the project data reliably, and Docker helped us set up the database environment consistently across team members' machines. We also used Bun and the project's full-stack development tools to run the frontend, backend, and database integration smoothly, which made development, testing, and system setup more organized.

6. Real-World Relevance

How does your project relate to real-world database applications?

Our project relates to real-world database applications because it models how an actual online learning platform stores, manages, and retrieves large amounts of connected data such as users, courses, lessons, purchases, reviews, and learning progress. Like real production systems, it requires a well-structured database to maintain consistency, support multiple features at once, and provide efficient access to information for both users and administrators. This makes the project a practical example of how database systems are used to power modern web applications.

7. Personal Improvement Reflection

If you were to do this project again, what would you do differently?

If I were to do this project again, I would spend more time at the beginning refining the database design and planning the relationships between features before implementation, because that would make later development smoother and reduce the need for changes along the way. I would also focus more on query optimization, indexing, and testing more edge cases earlier so the system would be more efficient and reliable from the start.

8. Final Thoughts

Any closing comments on your learning, project experience, or course.

This project gave me a much better understanding of how database concepts are applied in a real system, not just in theory. It helped me see the importance of good schema design, data relationships, and structured backend integration in building a reliable application. Overall, the experience was challenging but very valuable, and it made the course more practical and meaningful.